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1. Sodium fast reactors utilize fast spectrum neutrons that significantly affect safety aspects including shielding, operating temperature, and other safety metrics. The fast neutrons have a higher potential for elastic collisions wherein the neutrons don't always interact in a way yielding fission. This actually drives more implicit safety functions than those seen in Light-Water reactors, and decreases the shielding requirements. SFRs also have improved heat transfer between the rods and the coolant, yielding more efficient heat transfer while decreasing the fuel centerline temperature.

2. Uranium metal itself yields very anisotropic behavior with significant swelling. The orthorhombic microstructure contributes to significant swelling with increased temperature, which is far in excess to swelling observed in alloyed systems. This makes it challenging for designing of fuel assemblies since the fuel slug size must account for excessive anisotropic swelling at not only high temperature, but also high burnup. An alloyed uranium metal however, such as U-10Zr or U-Pu-Zr, yields more favorable swelling and fuel performance metrics while retaining significantly increased thermal conductivity compared to ceramic fuels.

3. Metallic fuel systems require a lower smear density compared to ceramic fuels because fuel swelling is significantly higher. This greatly increases potential for FCMI and FCCI. Fission gas release to the sodium bonding layer and plenum free surface is also increased compared to ceramic fuels. Linear growth in the fuel axial direction can occur and compress the plenum region prior to fuel bonding to the cladding, driving a larger plenum volume to reduce pressure increases in the plenum volume.

4. Constituent redistribution in metallic fuels occurs due to thermal and solubility gradients within the fuel. As fission gas forms and voids form, fission gas constituents can be transported to other areas of the fuel. This can form regions of different constituents depending on the thermal profile of the fuel. At the outer radius of the fuel there can be higher concentrations of cladding constituents, however, the fission product constituents can relocate to the higher temperature inner radius of the fuel (such as Cs), especially the soluble fission products. This phenomenon significantly

4. (continued) the thermal properties of the fuel and the fuel porosity as burnup increases.

5. thermal conductivity varies with burnup. At lower burnup, porosity forms and sodium from the sodium bonding layer can fill pores near the cladding. This enhances the thermal conductivity, improving heat transfer. Eventually at higher burnups, the porosity increases substantially, swelling increases, and more of the sodium is driven towards the plenum volume. the increased porosity, especially at the fuel centerline, will decrease the thermal conductivity.

6. FCCI is fuel - cladding chemical interaction, and it affects the metallic fuel assembly in at least two ways: cladding wastage and fuel eutectic formation at the fuel-cladding interface. As the fuel begins interacting with the cladding at higher burnup, chemical diffusion of cladding species, primarily Fe, will react with uranium and also Zr inside the fuel. This can more or less "erode" the thickness of the cladding, but it also forms a U-Fe or U-Fe-Zr eutectic phase that decreases the fuel melting temperature. This can ultimately lead to bulk fuel failure at higher burnup. FCCI layer generally is formed at higher burnup.

7. In MOX fuels, restructuring occurs as porous pathways/voids form when fission product transport occurs within the fuel. Under a large thermal or solubility gradient, fission products can leave a "trailing" porous structure allowing rapid transport of gaseous species and easier movement of non-soluble species within the fuel matrix/microstructure. Constituents within the fuel can shift around, leading to high concentrations of varying elements, resulting in a non-homogeneous fuel structure.

8. Pu diffusion and agglomerates form or are transported toward central voids in the fuel structure, leading to high Pu concentrations in these regions. Oxygen can be transported throughout the structure, not uniformly, but also in less concentrated pockets. The oxygen can be reabsorbed by non-alloyed metal constituents or other non-soluble constituents within the swelled region. The oxygen can also contribute to JOG formation.

9. JOG is joint oxide gain. Within the region between the fuel slug and cladding inside diameter, generated fission products can combine with oxygen and form a solid layer between the cladding and the fuel. Free oxygen left in the system after metallic agglomerates (such as Pu) are transported to voids can react with fission products, generating an additional layer between the fuel and cladding. JOG can increase the thermal conductivity and promote release of soluable fission gases to the plenum volume. This can relieve stress in the fueled region.

10. Sodium corrosion happens in two primary ways: 1) absorption of alloying elements from structural materials up to the solubility limit, and 2) reaction of sodium with impurities within the structural material. Stainless steels are common materials to use in SFRs, especially austenitic stainless steels as the solubility limit is quiet small up to 550 °C. However, nickle can be diffused into the sodium easier than other alloying elements, especially at higher temperatures.